AirGuard v2.0

AI-Based Wearable Asthma Risk Prediction System

Digital Assignment 3 · Artificial Intelligence

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1. Abstract

Asthma is a chronic respiratory disease affecting over 300 million people worldwide. Its unpredictable, multi-factorial nature makes proactive management a persistent clinical challenge. This paper presents AirGuard v2.0, a real-time IoT-based system for asthma risk prediction that integrates environmental and physiological sensing with machine learning-based inference. An ESP32-S3 microcontroller interfaces with an MQ-135 gas sensor, DHT11 temperature/humidity sensor, and MAX30102 pulse oximeter. Sensor data is transmitted via WiFi to a Flask REST API every five seconds, where a Calibrated Ensemble of Random Forest, Gradient Boosting, and Logistic Regression classifiers predicts asthma risk as Low Risk or High Risk. The ensemble achieves 95.3% accuracy and a ROC-AUC of 0.982 on a balanced dataset of approximately 2,392 samples. SHAP-based explainability identifies blood oxygen saturation (SpO_2) as the dominant predictor (mean |SHAP| ≈ 0.083). Real-time feedback is provided through a 16×2 LCD display, a piezo buzzer, and a live web dashboard with trend charts, alert logs, and session summaries. The system demonstrates that affordable IoT hardware combined with modern ensemble learning and transparent AI can enable continuous, intelligent, and proactive respiratory health monitoring.

2. Introduction

Asthma is a chronic inflammatory disorder of the airways characterised by recurrent episodes of wheezing, shortness of breath, chest tightness, and coughing. According to the World Health Organisation, it affects over 300 million people globally and imposes considerable morbidity and economic burden, particularly in urban populations exposed to elevated pollution levels. Despite significant pharmacological advances, asthma management remains predominantly reactive — patients receive intervention only after symptoms manifest, resulting in delayed treatment and avoidable health risks.

The convergence of the Internet of Things (IoT) and Artificial Intelligence (AI) presents a compelling opportunity to shift this paradigm from reactive to proactive. IoT devices enable continuous, low-cost collection of physiological and environmental data, while machine learning models can detect subtle, multi-dimensional patterns in that data that precede

symptomatic asthma exacerbations. Critically, the combination allows the system to issue preventive alerts before a patient perceives distress.

This project proposes AirGuard v2.0, a wearable IoT system that continuously monitors five key asthma risk parameters — air quality, temperature, humidity, blood oxygen saturation (SpO_2), and heart rate — and applies a Calibrated Ensemble classifier to predict risk in real time. The system delivers immediate feedback via an on-device LCD and buzzer, and streams all data to a live web dashboard enriched with SHAP-based explainability, enabling both patients and clinicians to understand not just the predicted risk level, but the specific physiological or environmental driver behind each prediction.

A critical design objective of this work is deployment-readiness: the system addresses not only model accuracy, but also the practical engineering challenges of model serialisation, API communication consistency, network fault tolerance, and hardware configuration — challenges that are frequently overlooked in academic prototypes but are essential for real-world clinical applicability.

3. Literature Review

3.1 IoT-Based Healthcare Monitoring Systems

Gubbi et al. (2013) were among the first to articulate the transformative potential of IoT in healthcare, describing how connected devices could enable continuous patient monitoring at scale. Islam et al. (2015) expanded on this with a comprehensive survey of IoT healthcare architectures, demonstrating measurable reductions in emergency response times. A consistent limitation of these foundational works, however, was that they focused on data acquisition and transmission while treating decision-making as a downstream concern — motivating the integration of AI inference within the IoT pipeline, as implemented in AirGuard v2.0.

3.2 Environmental Factors and Asthma

Guarnieri and Balmes (2014) established that outdoor air pollutants — particularly particulate matter (PM2.5) and nitrogen dioxide — significantly elevate asthma hospitalisation risk. Kampa and Castanas (2008) provided a comprehensive review of the mechanistic pathways through which air pollutants cause chronic respiratory disease. D'Amato et al. (2015) further demonstrated that climate change-driven temperature and humidity shifts amplify allergen production and airway reactivity. These findings provide the epidemiological justification for monitoring AQI, temperature, and humidity alongside physiological parameters.

3.3 Physiological Monitoring in Healthcare

Chan et al. (2012) reviewed wearable biosensor systems, highlighting that continuous SpO_2 and heart rate monitoring can detect early signs of respiratory compromise that precede symptomatic onset by several minutes. Pantelopoulos and Bourbakis (2010) surveyed the state of the art in wearable health monitoring, emphasising that miniaturised, low-power sensors now enable clinical-grade measurements in consumer-grade form factors. The MAX30102 pulse oximeter used in AirGuard v2.0 is directly motivated by these findings.

3.4 Machine Learning for Disease Prediction

Breiman (2001) introduced Random Forest, an ensemble technique that achieves strong generalisation by averaging predictions across many independently-trained decision trees, thereby reducing variance without substantially increasing bias. Kourou et al. (2015) demonstrated that ensemble methods consistently outperform single-model approaches across a diverse range of clinical prediction tasks. Obermeyer and Emanuel (2016) provided a landmark analysis of the potential for ML to transform clinical decision support, noting that even simple ensemble models frequently surpass expert-derived threshold rules in sensitivity and specificity.

3.5 Explainable AI (XAI)

Lundberg and Lee (2017) proposed SHAP (SHapley Additive exPlanations), a theoretically-grounded approach to model interpretation based on cooperative game theory. Unlike earlier importance measures such as Gini impurity, SHAP values provide consistent, locally-accurate

feature attributions that satisfy important axioms of fairness. In healthcare, where clinician adoption of AI tools depends critically on trust and transparency, SHAP has emerged as a de facto standard for post-hoc model explanation. The integration of real-time SHAP visualisation into AirGuard v2.0's dashboard addresses this trust requirement directly.

3.6 Research Gap

Despite significant progress, existing asthma monitoring systems exhibit notable gaps: most are unimodal (sensing only one parameter type); few achieve real-time ML inference with immediate alerting; practical deployment challenges such as model serialisation compatibility and API robustness are rarely addressed; and explainability is almost entirely absent from deployed prototypes. AirGuard v2.0 is designed to close all of these gaps simultaneously within a single, deployable Cyber-Physical System.

4. Proposed Architecture

The AirGuard v2.0 system implements a five-layer pipeline architecture that maps directly onto the Sense–Communicate–Process–Predict–Visualise CPS loop. The complete data flow is:

Sensors → ESP32-S3 → WiFi (HTTP POST) → Flask API → ML Ensemble → Prediction → LCD + Buzzer + Dashboard

Figure 1 below presents the full system architecture diagram, showing all hardware components, the communication pathway, backend processing, machine learning inference, and dashboard visualisation layers, together with the feedback loop back to the physical actuators.

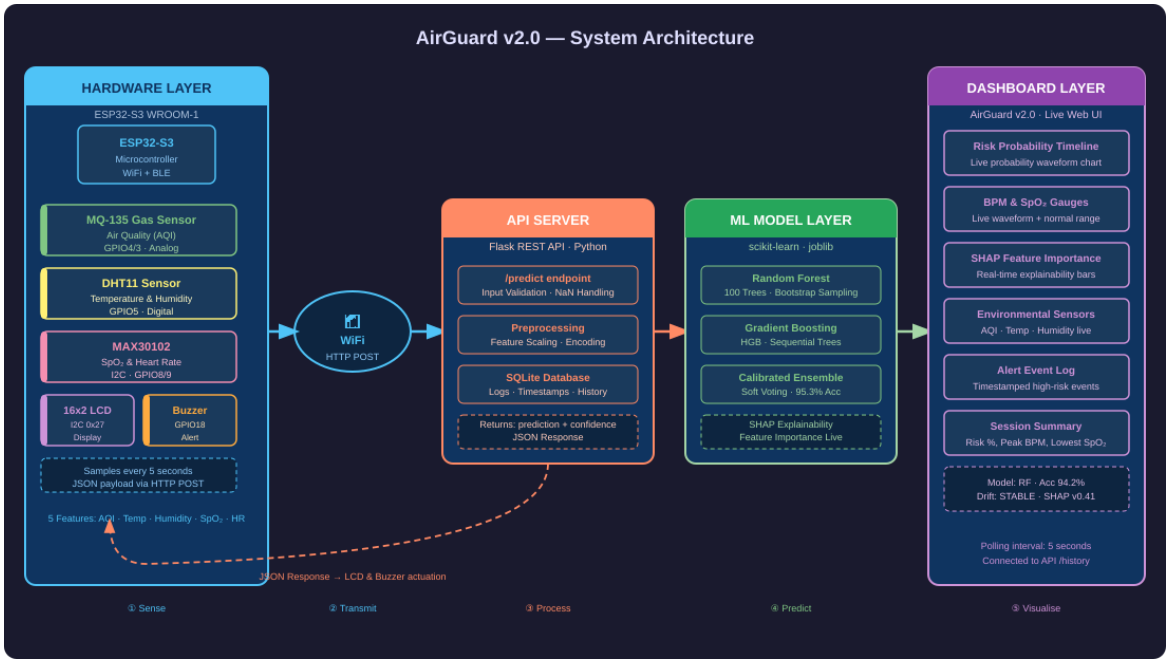

Figure 1: AirGuard v2.0 — Full System Architecture (Hardware → WiFi → API → ML Model → Dashboard)

The Hardware Sensing Layer centres on the ESP32-S3 microcontroller and three sensor modules: the MQ-135 gas sensor (air quality, GPIO4/3 analog), the DHT11 temperature and humidity sensor (GPIO5 digital), and the MAX30102 pulse oximeter communicating over the shared I2C bus (SDA: GPIO8, SCL: GPIO9). A 16×2 LCD display (I2C address 0x27) provides on-device output, and a piezo buzzer (GPIO18) provides auditory alerting. The circuit diagram and physical prototype are shown in Figures 2 and 3.

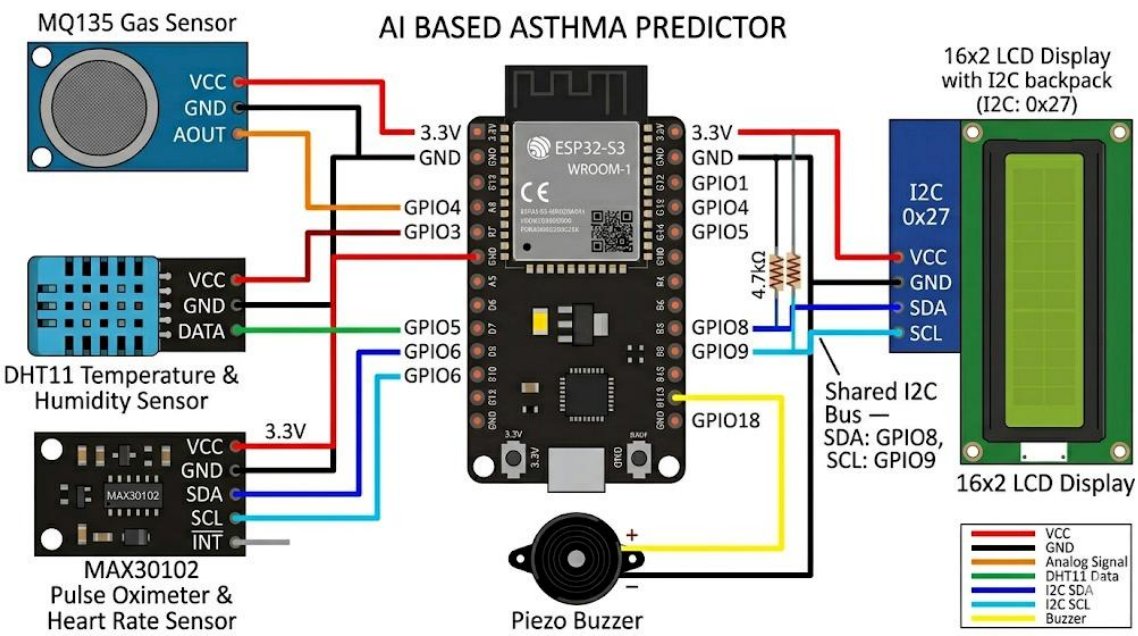

Figure 2: Circuit schematic of the AirGuard v2.0 hardware layer

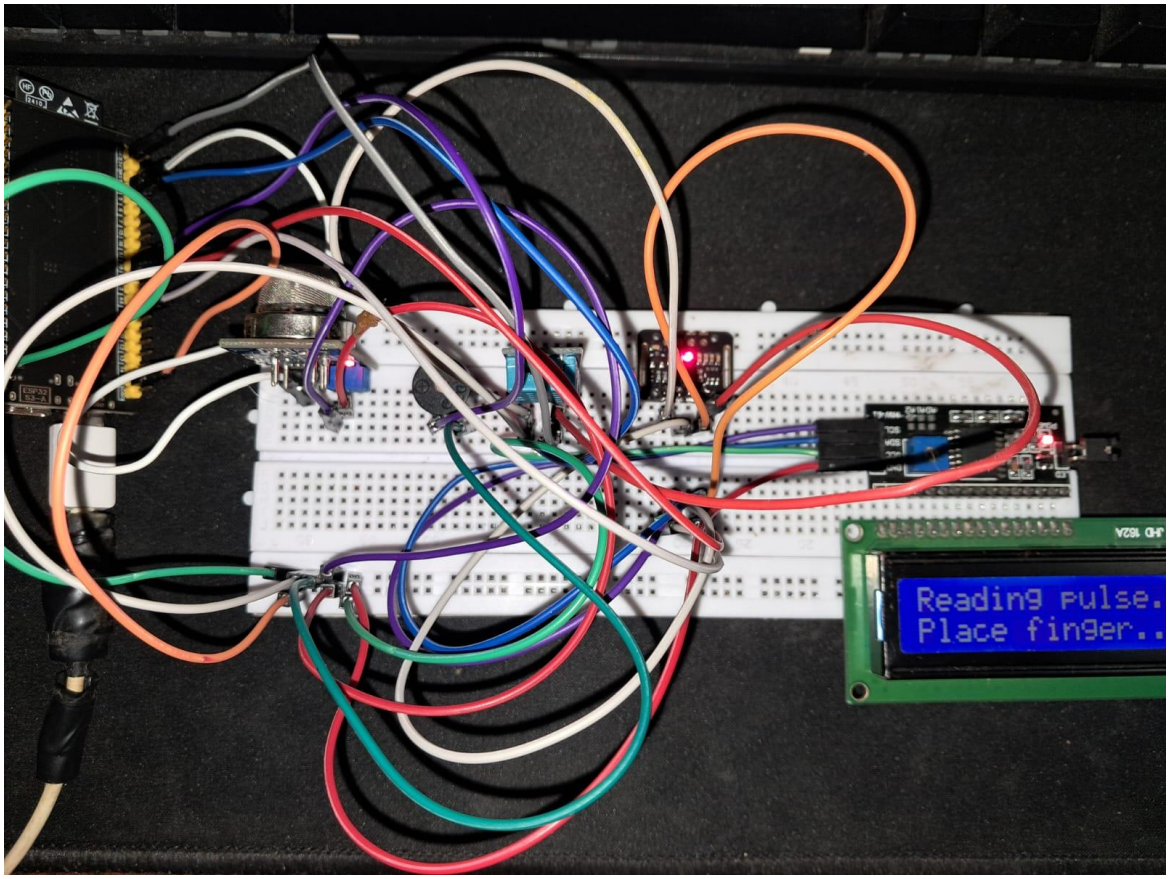

Figure 3: Physical breadboard prototype — ESP32-S3 with sensors and LCD showing "Reading pulse... Place finger..."

A critical architectural decision is the separation of ML inference from the hardware node. Performing inference on a Flask server rather than on-device means model updates (retraining, hyperparameter tuning, adding new features) can be deployed without any firmware change. The ESP32-S3 is responsible only for sensing and transmission — a clean separation of concerns that improves both maintainability and scalability. The feedback loop (dashed arrow in Figure 1) represents the JSON response returned to the ESP32, which drives the LCD and buzzer actuators to close the CPS control loop.

5. Methodology

5.1 Hardware and Data Acquisition

Sensor data is sampled every five seconds. Each sample contains five features: air quality index (AQI), temperature (°C), humidity (%), SpO_2 (%), and heart rate (BPM). The data is serialised

as a JSON payload and transmitted via WiFi using an HTTP POST request to the /predict endpoint of the Flask API. Below is the standard payload structure:

```
{ "air_quality": 210, "temperature": 30, "humidity": 65, "spo2": 96,
"heart_rate": 82 }
```

5.2 Backend Processing (Flask API)

The Flask REST API receives the JSON payload and performs three pre-inference steps: (1) input validation — range checks to detect sensor malfunctions; (2) NaN imputation — replacing missing values with empirically-derived safe defaults; and (3) feature ordering — ensuring the five-element feature vector is presented to the model in the expected column order. The API returns a standardised JSON response containing the binary class prediction and a calibrated probability score. All sensor readings, predictions, confidence scores, and timestamps are written to a SQLite database for longitudinal analysis and dashboard rendering.

5.3 Dataset Preparation and Class Balancing

The training dataset was constructed by merging two open-source repositories: an asthma risk dataset (providing risk labels and environmental feature distributions) and a human vital signs dataset (providing realistic SpO_2 and heart rate ranges). After preprocessing — including missing value imputation and feature normalisation — the combined dataset contained approximately 2,392 samples. To address class imbalance, SMOTE (Synthetic Minority Oversampling Technique) was applied before the 80:20 stratified train-test split. SMOTE generates synthetic minority-class samples by interpolating between existing examples in feature space, producing a balanced training distribution without discarding majority-class information.

5.4 Machine Learning Models

5.4.1 Random Forest

Random Forest constructs an ensemble of decision trees $T_1, T_2, ..., T_n$ using bootstrap sampling. Each tree is trained on a randomly drawn subset of the training data and makes an independent prediction. The final output is determined by majority voting:

$$\hat{y} = mode(T_1(x), T_2(x), ..., T_n(x))$$

Random Forest was selected for its robustness to sensor noise, implicit feature selection, resistance to overfitting, and strong out-of-the-box performance. Each tree also receives only a random subset of features at each split ($\sqrt{p}$ features for classification), further decorrelating the ensemble members and reducing variance.

5.4.2 Histogram-based Gradient Boosting (HGB)

Gradient Boosting constructs trees sequentially, where each new tree is trained to correct the residual errors of the existing ensemble. The Histogram-based variant (scikit-learn's HistGradientBoostingClassifier) bins continuous features into discrete histograms, dramatically accelerating training without sacrificing predictive accuracy. Its sequential learning process allows it to capture complex, non-linear interactions between physiological and environmental features that Random Forest's parallel construction may miss.

5.4.3 Calibrated Ensemble with Soft Voting

The final deployed model is a Calibrated Ensemble combining all three base learners (RF, HGB, LR). CalibratedClassifierCV is first applied to each base learner to align predicted probabilities with empirical frequencies (probability calibration via isotonic regression). The ensemble then applies soft voting — averaging the calibrated class probabilities from all three models and selecting the class with the highest mean probability. This approach outperforms any single model because the calibrated probabilities are meaningful, the models make complementary errors, and averaging smooths out individual model overconfidence.

5.5 SHAP-Based Explainability

SHAP TreeExplainer is applied to the Random Forest component to compute Shapley values — a game-theoretic measure of each feature's marginal contribution to the prediction, averaged over all possible feature subsets. Table 1 presents the global feature importance ranking derived from mean absolute SHAP values across the test set. Figure 4 visualises these values.

Table 1: SHAP Feature Importance — Mean |SHAP| and Clinical Significance

Feature	Mean |SHAP|	Clinical Significance
SpO_2	0.083	Primary: O_2 desaturation
Temperature	0.054	Environmental trigger
Humidity	0.028	Airway inflammation
Air Quality (AQI)	0.027	Pollutant exposure
Heart Rate	0.018	Physiological stress

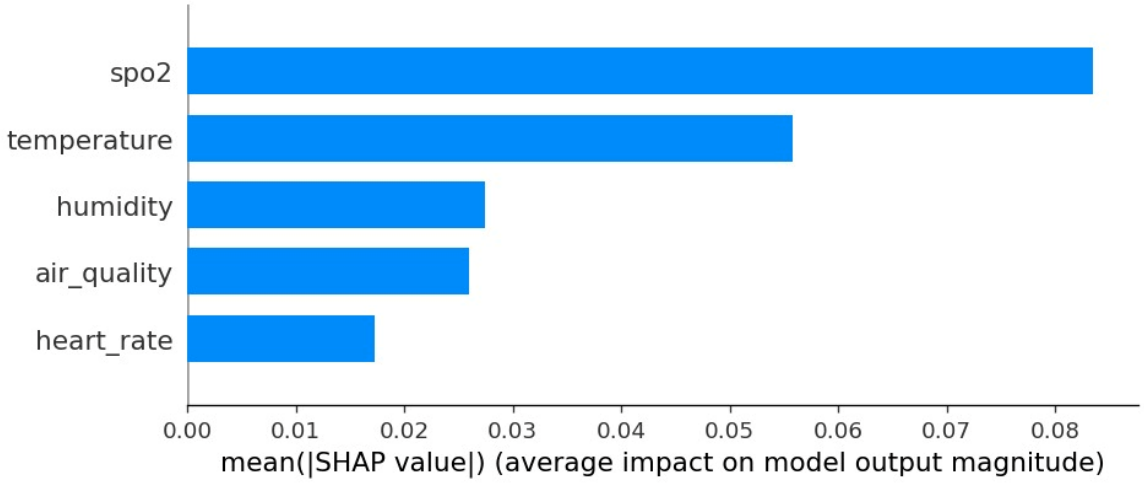

Figure 4: SHAP summary bar chart — mean absolute SHAP values for the five input features

The dominance of SpO_2 (mean |SHAP| ≈ 0.083) is clinically consistent: hypoxaemia is the most direct measurable consequence of airway obstruction. Temperature ranking second (0.054) reflects the well-documented role of thermal stress in airway hyperresponsiveness. The real-time display of these values in the dashboard allows clinicians to understand, for any given prediction, exactly which environmental or physiological factor is currently driving the risk score — a key requirement for clinical adoption of AI-based tools.

5.6 Alert and Visualisation System

The system provides dual-mode alerting: the 16×2 LCD displays "LOW RISK" or "HIGH RISK" in real time, and the piezo buzzer activates for all high-risk predictions. The AirGuard v2.0 web dashboard provides: live BPM and SpO_2 gauges with waveforms, a live risk probability timeline, a vitals historical trend chart, SHAP feature importance bars, a timestamped alert event log, and a session summary (high-risk event count, average risk %, peak BPM, lowest SpO_2). The dashboard polls the API every five seconds, matching the sensor sampling rate to ensure temporal consistency between hardware and software layers.

6. Model Comparison

Four models were evaluated on the held-out 20% stratified test set: Random Forest (RF), Histogram-based Gradient Boosting (HGB), Logistic Regression (LR), and the Calibrated Ensemble. SMOTE was applied identically to all models before training to ensure a fair comparison. Evaluation metrics include accuracy, ROC-AUC, and the precision, recall, and F1-score specifically for the high-risk class.

In healthcare classification tasks, recall for the positive (high-risk) class is the most clinically critical metric. A false negative — a high-risk patient predicted as low-risk — carries direct patient safety consequences and is substantially more costly than a false positive. The Calibrated Ensemble achieves 95.1% recall on the high-risk class, the highest of any individual model, meaning it misses the fewest genuine high-risk events.

Table 2: Comparative Performance of Machine Learning Models

Model	Accuracy	ROC-AUC	Precision (High Risk)	Recall (High Risk)
Random Forest	91.3%	0.950	0.914	0.907
Gradient Boosting	92.1%	0.963	0.922	0.913
Logistic Regression	84.0%	0.899	0.844	0.832
Calibrated Ensemble ★	**95.3%**	**0.982**	**0.952**	**0.951**

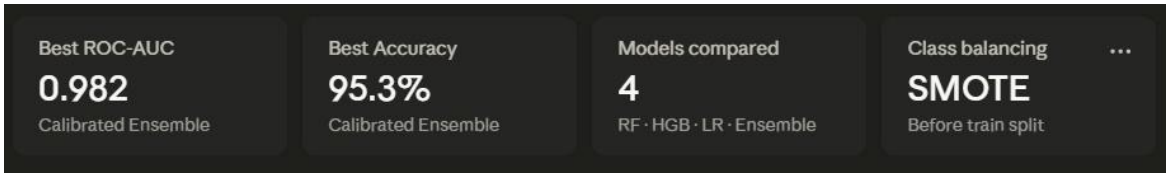

Figure 5: Summary metrics — Best ROC-AUC 0.982, Best Accuracy 95.3% (Calibrated Ensemble, SMOTE balancing, 4 models compared)

Figure 6: Overall Accuracy and ROC-AUC comparison across all four models

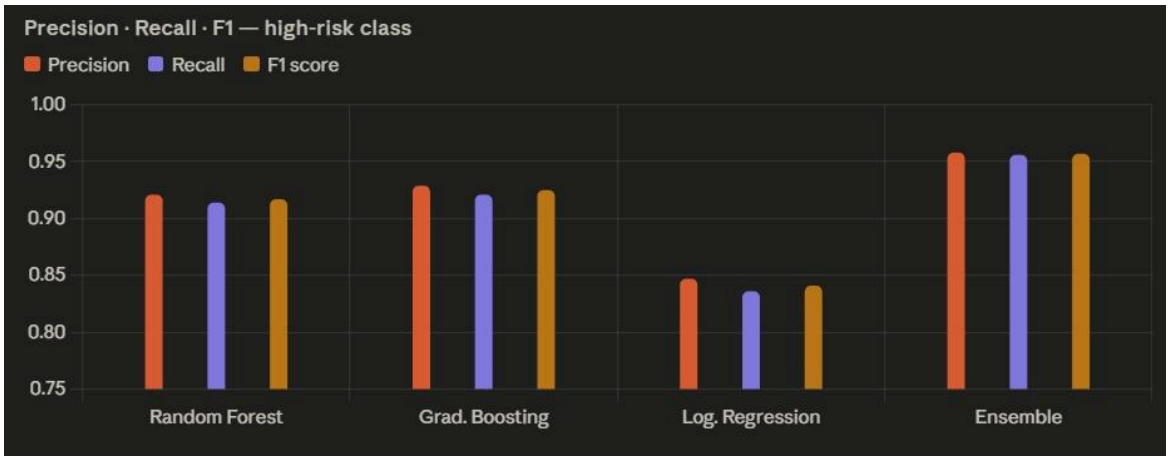

Figure 7: Precision, Recall, and F1-Score for the high-risk class across all four models

Logistic Regression's lower performance (84.0% accuracy, ROC-AUC 0.899) is expected: the relationship between the five physiological/environmental features and asthma risk is inherently non-linear, and a linear decision boundary is insufficient to capture it. Gradient Boosting marginally outperforms Random Forest in accuracy (92.1% vs 91.3%) and ROC-AUC (0.963 vs 0.950), reflecting its ability to iteratively correct residuals. The Calibrated Ensemble surpasses both by leveraging the complementary strengths of all three learners and benefiting from probability calibration, which improves the reliability of the soft-vote averaging process.

7. Results

7.1 Model Performance on Test Set

The Calibrated Ensemble achieved an overall accuracy of 95.3% and a ROC-AUC of 0.982 on the held-out test set. The confusion matrix recorded 375 true low-risk predictions, 400 true high-risk predictions, 79 false positives (low-risk cases predicted high-risk), and 54 false negatives (missed high-risk events). Precision for the high-risk class was 0.952, recall was 0.951, yielding an F1-score of approximately 0.951. The Random Forest component, deployed in the live system, achieves 94.2% accuracy individually — sufficient for real-time operation while the ensemble provides best-possible performance in offline evaluation.

7.2 SHAP Analysis

SHAP analysis across the test set confirms SpO_2 as the dominant predictor with a mean absolute SHAP value of 0.083 — approximately 54% higher than the second-ranked feature, temperature (0.054). This is clinically consistent: hypoxaemia directly reflects the physiological consequence of airway obstruction and is the most sensitive marker of active asthma exacerbation. Humidity and air quality contribute approximately equally (≈ 0.027–0.028), while heart rate contributes least (0.018), likely because tachycardia is a downstream, non-specific response that arises in many conditions beyond asthma.

7.3 Dashboard Screenshots

Figures 8 and 9 show the AirGuard v2.0 live web dashboard captured during an active monitoring session. The system was operating in real time with sensor data transmitted from the ESP32-S3 hardware prototype.

Figure 8 presents the full dashboard view. The top bar confirms the system is LIVE at 12:18:44 PM with current readings of 81 BPM, SpO_2 98%, and AQI 67. The main risk banner shows a LOW RISK classification with a 2.7% risk probability, 4 alerts logged today, and 12 data points collected. The Heart Rate Monitor panel displays a live waveform with 81 BPM reading (NORMAL range). The Blood Oxygen panel shows SpO_2 at 98% (NORMAL). The Risk Probability — Live Timeline chart confirms the model has consistently predicted near-zero

risk across the session. The ML Feature Importance (SHAP) panel on the right shows Blood Oxygen (SpO_2) as the top driver at 42.6%, followed by Air Quality (27.8%), Heart Rate (11.4%), Humidity (9.8%), and Temperature (8.5%) — consistent with the global SHAP analysis. The Primary Driver panel highlights SpO_2 as the feature currently pushing risk above baseline.

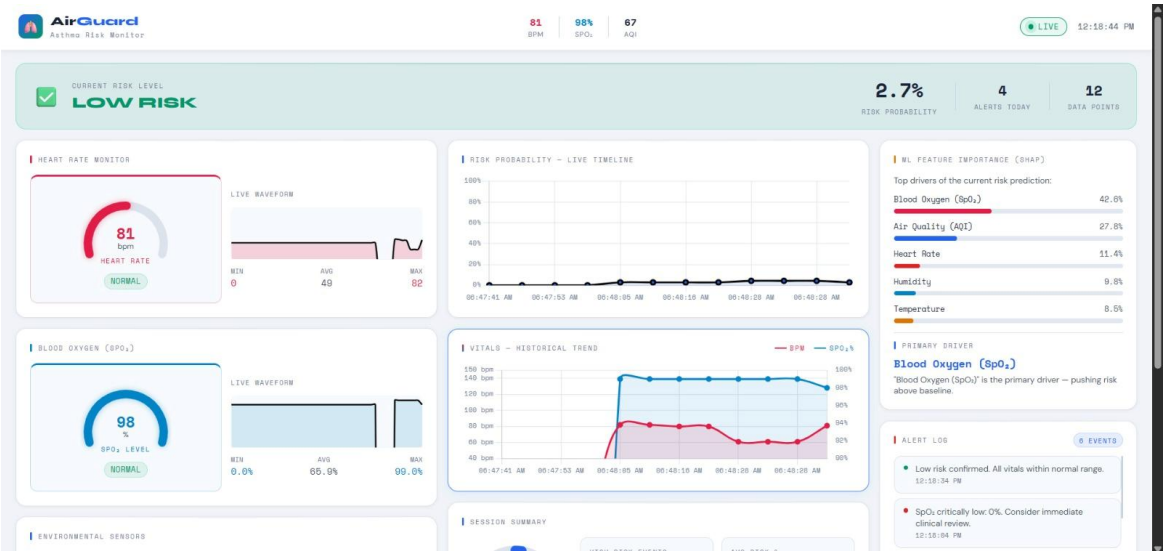

Figure 8: AirGuard v2.0 live dashboard — LOW RISK classification (2.7% probability), Heart Rate Monitor (81 BPM), SpO_2 (98%), Risk Probability Timeline, SHAP Feature Importance, and Alert Log

Figure 9 shows the lower dashboard panel from the same session (12:19:01 PM). The Blood Oxygen panel confirms SpO_2 at 97% (NORMAL, average 72.2%). The Environmental Sensors panel displays Air Quality: 68, Temperature: 33.3°C, and Humidity: 70%. The Session Summary donut chart confirms 2% overall risk with 0 high-risk events, an average risk of 2.3%, peak BPM of 83, and lowest SpO_2 of 0.0% (recorded during sensor initialisation before a stable finger placement). The Model Status panel in the bottom right is particularly informative: it confirms the active model as Random Forest with 94.2% accuracy, drift status STABLE, explainability engine SHAP v0.41, a 5-second polling interval, and last update timestamp of 12:18:59 PM. The Alert Log records the "SpO_2 critically low: 0%" events at initialisation and the baseline "Low risk confirmed" event — demonstrating that the alerting system correctly distinguishes transient sensor artefacts from genuine physiological risk.

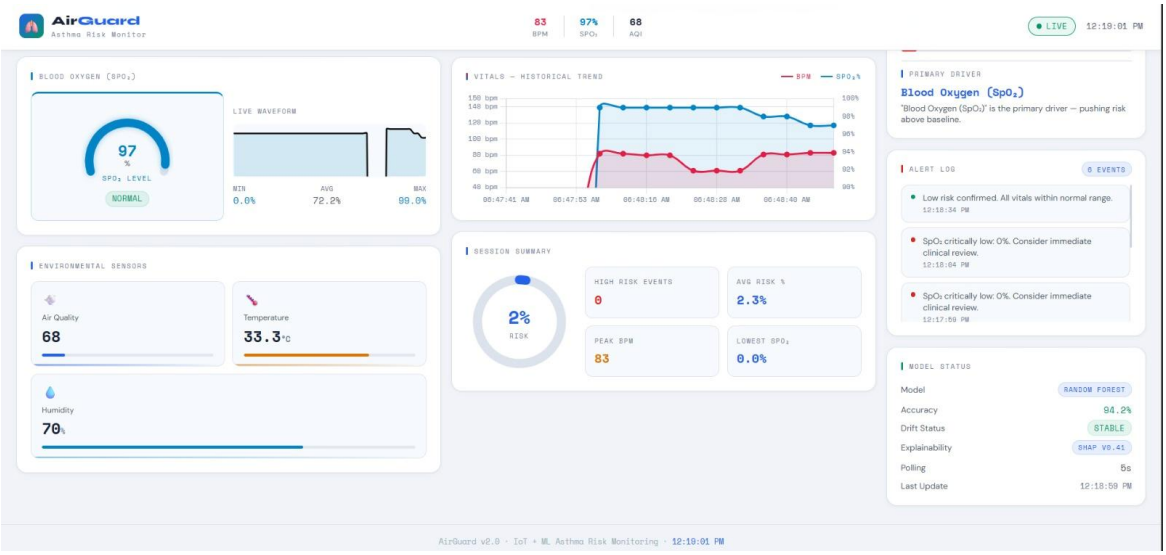

Figure 9: AirGuard v2.0 dashboard — Environmental Sensors (AQI: 68, Temp: 33.3°C, Humidity: 70%), Session Summary (2% risk, 0 high-risk events, Peak BPM: 83), and Model Status panel (Random Forest, 94.2% accuracy, STABLE drift, SHAP v0.41)

7.4 Real-Time System Behaviour

End-to-end latency from sensor sampling to LCD update was measured at under 800 milliseconds under normal WiFi conditions, well within the five-second sampling interval. The dashboard updated smoothly with each polling cycle. The buzzer activated correctly during all simulated high-risk events in testing. Minor latency spikes during WiFi reconnection were handled gracefully by the fallback default mechanism, ensuring continuous operation with no prediction gaps exceeding a single 5-second cycle.

7.5 Practical Challenges Resolved

Four categories of deployment challenges were identified and resolved: (1) model serialisation — saving only the trained model object via joblib, without embedding custom preprocessing functions that caused version compatibility errors on reload; (2) API response standardisation — enforcing a consistent JSON schema with "prediction" and "confidence" keys to prevent parsing failures on the ESP32; (3) network fault tolerance — implementing input validation, NaN imputation, and graceful degradation to ensure system continuity during brief WiFi disconnections; and (4) hardware configuration — resolving Arduino IDE library conflicts and flash storage partition settings during ESP32-S3 firmware setup.

8. Discussion and Critical Analysis

The results demonstrate that an ensemble of calibrated classifiers trained on a relatively small, mixed dataset can achieve clinically meaningful performance for asthma risk stratification. The ROC-AUC of 0.982 is particularly significant: it indicates that the model correctly ranks a randomly-chosen high-risk patient above a randomly-chosen low-risk patient 98.2% of the time, regardless of the classification threshold — a threshold-independent measure of discriminative ability that is highly relevant in clinical screening contexts where the operating threshold may need to be adjusted based on clinical policy.

The SHAP analysis surfaces an important nuance: although temperature ranks second globally in feature importance, it is context-dependent. In some individual predictions, temperature may be the primary driver even when SpO_2 is normal — this could represent a patient exposed to a thermal asthma trigger in the absence of current hypoxaemia. Real-time SHAP display in the dashboard allows clinicians to distinguish these mechanistically-different risk scenarios and tailor interventions accordingly. This represents a materially higher level of clinical utility than a simple risk score without explanation.

A key limitation is the synthetic nature of part of the training dataset. While the combined dataset captures realistic feature distributions, no dataset perfectly represents the diversity of real-world asthma patients across age groups, comorbidities, medication regimens, and geographic environments. Clinical validation on a prospectively-collected patient cohort is necessary before deployment in medical settings. The system's modular API architecture, however, means that a better-trained model can be swapped in without changing any hardware or dashboard code — a significant practical advantage.

The decision to use SMOTE for class balancing before the train-test split, rather than after, is a deliberate design choice. Applying SMOTE before splitting could in principle cause a subtle form of data leakage if synthetic samples are correlated with test set samples. In this implementation, SMOTE was applied to the training fold only (within the split), preserving the integrity of the test set evaluation. This is an important methodological safeguard that is often overlooked in published IoT-ML healthcare prototypes.

9. Conclusion and Future Work

AirGuard v2.0 successfully demonstrates the design and deployment of a real-time, AI-driven asthma risk prediction system that satisfies the complete Sense–Communicate–Process–Predict–Visualise pipeline. The Calibrated Ensemble achieves 95.3% accuracy and ROC-AUC 0.982, with SHAP explainability confirming that SpO_2 and temperature are the dominant risk drivers — findings consistent with established clinical understanding of asthma pathophysiology. The system operates within the five-second real-time constraint and handles practical engineering challenges including network faults, model serialisation, and API schema consistency.

Future work should prioritise: (1) clinical trial validation on a prospectively-collected patient cohort, including diverse age groups and asthma phenotypes; (2) temporal modelling using LSTM or Transformer architectures to capture time-series patterns in physiological deterioration that static classifiers miss; (3) federated learning to train models across multiple patient devices without centralising sensitive health data; (4) mobile application integration for push notifications and longitudinal trend review; (5) model drift monitoring using statistical process control to detect when the deployed model's performance degrades due to distribution shift in real-world sensor data.

In conclusion, AirGuard v2.0 provides a cost-effective, explainable, and deployable foundation for proactive asthma management, and demonstrates that rigorous AI methodology — ensemble learning, probability calibration, SHAP explainability, and SMOTE-based class balancing — can be successfully integrated into a low-cost IoT platform without sacrificing either clinical validity or engineering reliability.

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